

Data Science with Python – Module 1 Exercise

Hands-on Exercise No. 1

Course: Data Science with Python

Module: Python Basics & Data Analysis

Total Marks: 10

Due Date: 21/06/2026

Instructions

Prepare your solution in MS Word.

Submit in .doc or .docx format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Write Python scripts for basic data analysis.
 - Use Pandas for data manipulation.
 - Perform descriptive statistics on datasets.
 - Visualize data using Matplotlib or Seaborn.
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Problem Statement

You are working as a Data Analyst for a retail company. The client wants insights into customer purchase behavior using Python. Your task is to analyze a dataset and generate meaningful statistics and charts.

Task Requirements

Create a new Jupyter Notebook or Python script.

Import Pandas and load a dataset (for example, `sales.csv`).

Perform the following operations:

- Display the first 10 rows of the dataset.
- Show dataset shape and column names.

- Calculate the average purchase amount.
- Identify the top 5 customers by total purchase value.
- Handle missing values by either filling with mean or dropping rows.

Create visualizations using Matplotlib or Seaborn:

- Histogram of purchase amounts.
 - Bar chart of top 5 customers.
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Take Screenshots of

- Python code snippets.
 - Dataset output results.
 - Data visualizations (graphs/charts).
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Submission Guidelines

Upload your .docx file with code snippets, screenshots, charts, and explanations via LMS.